Assumption University
Vincent Mary School of Engineering, Science and Technology
Quiz1 CSX3007 (70 points) 2/2025 **SOLUTION**

Instructions: Calculators are not permitted, and this is a closed-book examination.

1. [**20 points**] **Number System**

1.1) (1 point) Show the 8-bit sign-magnitude representation of -6. **=1000 0110**

1.2) (1 point) Show the result of the following *hexadecimal subtraction*: 0x7E – 0x5B

= 0x23 = 00100011

1.3) (2 points) Show the decimal equivalent of the following 8-bit 2's complement binary numbers.

1.3.1. (1 point) 11110000_2 **-16**

1.3.2. (1 point) 11111000_2 **-8**

1.4) (2 points) Convert the following *unsigned binary* numbers to *decimal*:

1.4.1. (1 point) 00101101_2 **45**

1.4.2. (1 point) 00110101_2 **53**

1.5) (1 point) Convert the following *unsigned binary* number to *hexadecimal*:

1101101010111011_2 **0xDABB**

1.6) (1 point) Convert the following *hexadecimal* number to *decimal*: $A0F_{16}$ **2575**

1.7) (1 point) Repeat **question 1.6** but convert to an *unsigned binary* number.

101000001111

1.8) (1 point) Convert the following *hexadecimal* number to a *two's complement binary* number: $28BA_{16}$ **1101 0111 0100 0110**

1.9) (3 points) Answer the following:

1.9.1. (0.5 points) Convert 9057_{10} to BCD (Binary Coded Decimal).
1001000001010111

1.9.2. (0.5 points) Convert $01011001011110011000_{BCD}$ to decimal. **59798**

1.9.3. (3 points) Show the equivalent BCD results of the following decimal addition:
$78_{10} + 95_{10} = 173_{10}$

```
78+   --> 0111 1000 +
95    --> 1001 0101
---       ---------
173     1 0000 1101 +
        0 0000 0110
        -----------
        1 0001 0011+
        0 0110 0000
      -------------
    0001 0111 0011 --> 173
```

1.10) (1 point) Show the results of a one-bit *shift right* on the following 8-bit binary number: 01111011_2 **00111101**

1.11) (1 point) Show the results of a one-bit *rotate right* on the following 8-bit binary number: 10111001_2 **11011100**

1.12) (2 points) Check whether the following binary addition causes arithmetic flags such as *overflow*, *carry*, *sign*, and *zero*: $11111111_2 + 10000001_2$ = **10000000**

CF = 1, SF = 1, OF = 0, ZF = 0, PF =1 (if *even* parity = 0, and *odd* parity = 1)

1.13) (3 points) Convert **1.1001 × 2^18** into a 32-bit **IEEE** floating-point format.

0 (sign bit) 10010001 (8-bit biased exponent) 0001100……(23-bit mantissa)

2. [**34 points**] **Combinational Logic**

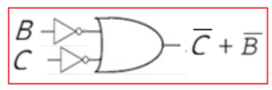
2.1) (6 marks) Consider the following truth table (where *A*, *B*, and *C* are **three** input variables and *Y* is an output variable):

A	***B***	***C***	***Y***
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

2.1.1. (2 points) Write a Boolean equation in *sum-of-products* canonical form.

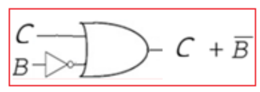
$Y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C}$

2.1.2. (4 points) Minimize the equation from **2.1.1** using a ***K-map*** and sketch its combinational logic circuit. $\bar{B} + \bar{C}$

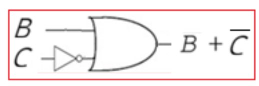


2.2) (4 points) Simplify the following logic equations and *sketch the circuits*:

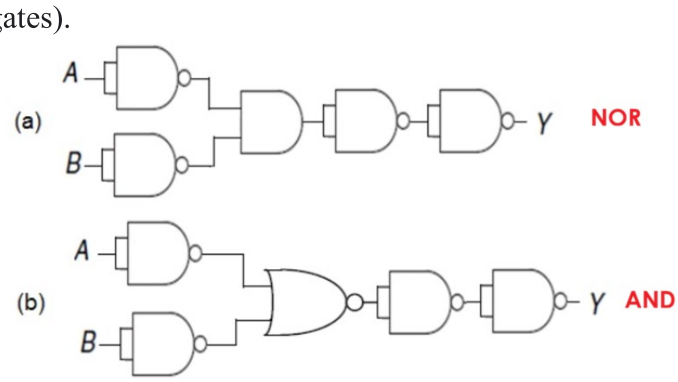
2.2.1. (2 points) $Y = A\bar{B} + \bar{A}\,\bar{B}\,\bar{C} + ABC + \bar{A}C = \bar{B} + C$



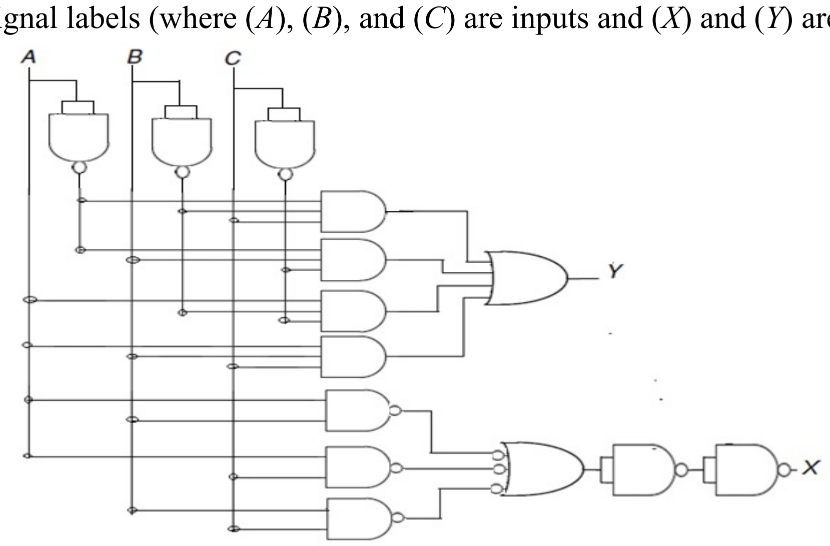
2.2.2. (2 points) $Y = \overline{(A + \bar{B} + C)} + (\overline{\bar{B}C}) + ABC = B + \bar{C}$



2.3) (3 points) There are two **2-input logic gate** functions shown in **Figures (a) and (b)** below (where A, B, and are inputs and outputs, respectively). Identify the logic gates (name the logic gates).

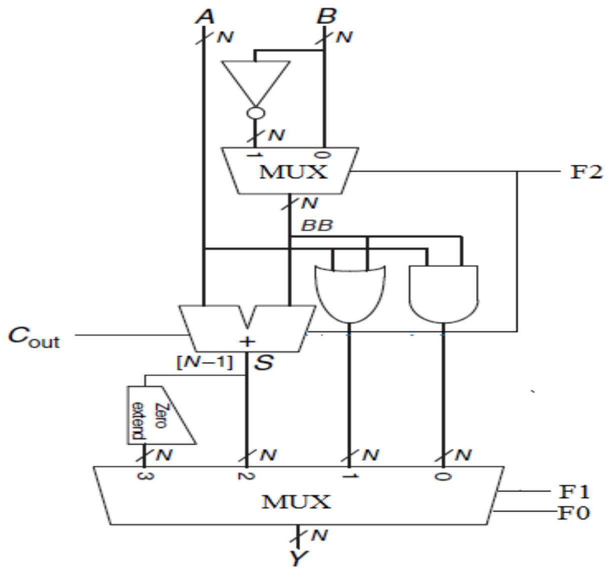


2.4) (3 points) Identify the circuit schematic shown in the **Figure** below by applying its input and output signal labels (where (A), (B), and (C) are inputs and (X) and (Y) are outputs).



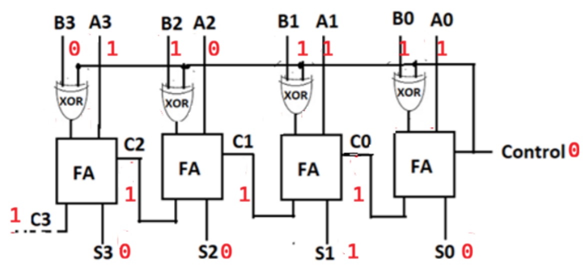
Full Adder: *A*, *B*, and *C* are inputs, *Y* is the sum, and *X* is the carry$_{out}$.

2.5) (3 points) Answer the following based on the circuit schematic shown in the **Figure** below (where ***A*** and ***B*** are two N-bit binary inputs, and C_{out} is the carryout bit): a) Identify the function of the given circuit. **ALU** b) Find the output, Y, when $F_2 = 0$, $F_1 = 1$, and $F_0 = 0$. **Addition (A+B)** c) Find the output, Y, when $F_2 = 1$, $F_1 = 1$, and $F_0 = 0$. **Subtraction (A - B)**

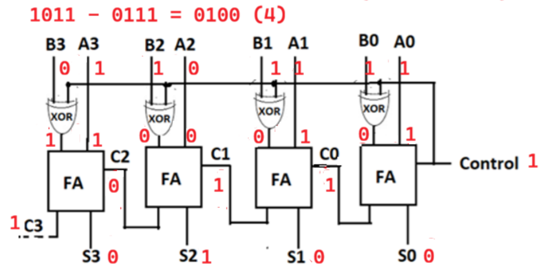


2.6) (4 points) Show the circuit schematic of a **full adder** with the help of its truth table. Show how a 4-bit **subtraction** is possible with the full adders (use the 4-bit binary values 1011_2 and 0111_2 for its addition and subtraction demonstration).

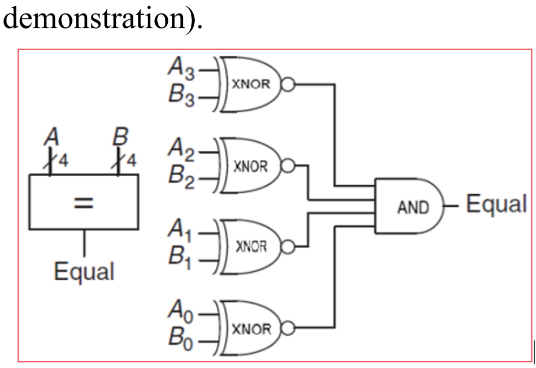
Full Adders for Addition (control = 0):
1011 + 0111 = 0010



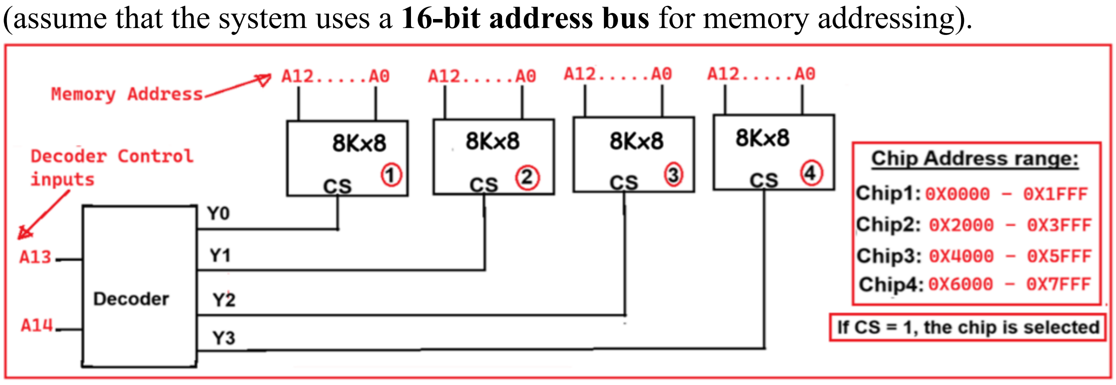
Full Adders for Subtraction (Control = 1):
1011 - 0111 = 0100 (4)



2.7) (3 points) Show the circuit schematics of a **4-bit equality** comparator (use any 4-bit binary values for its demonstration).



2.8) (4 points) Show how a 4-line Decoder acts as a memory chip selector for a memory board with four **8KB (8K x 8)** chips. Find the **address range** of each memory chip (assume that the system uses a **16-bit address bus** for memory addressing).



2.9) (4 points) Assume that a computer system has a **2GB** main memory (organization: **2G x 8**). Answer the following based on the main memory:

2.9.1. (1.5 points) How many *control bits* are in the address decoder of the memory? **31**

2.9.2. (1.5 points) How many *output bits* are in the address decoder of the memory? **2G**

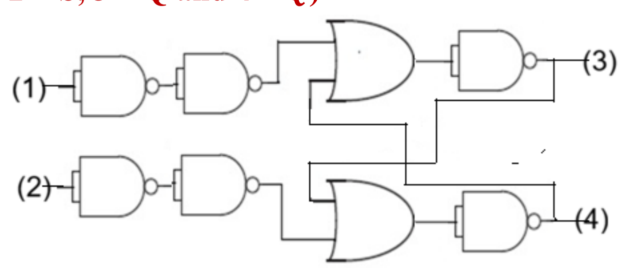
2.9.3. (1 point) How many *data bits* are needed to access the memory? **8**

3. [16 points] Sequential Logic

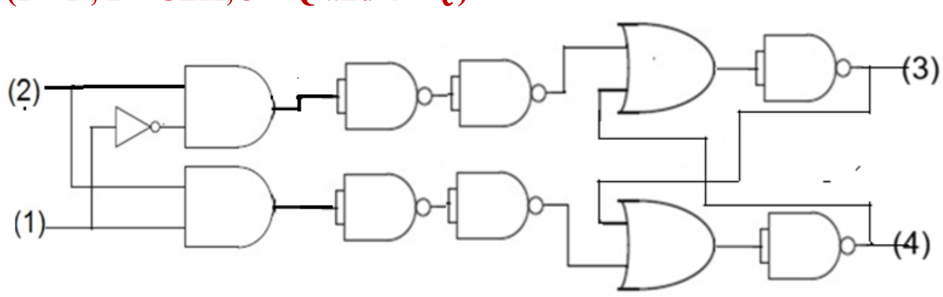
3.1) (1 point) Why are DRAM (Dynamic RAM) and SRAM (Static RAM) volatile? **DRAM (Main memory) and SRAM (cache) support the CPU in instruction execution. Therefore, after each execution, both of these memories must be cleared for subsequent executions (they serve as an instruction/data buffer for the CPU).**

3.2) (1 point) Is it possible to construct an EEPROM with DRAM technology? Why? **No, it is not possible to construct an EEPROM using DRAM technology, because EEPROMs are non-volatile semiconductor memories. In contrast, DRAM storage elements are capacitors and are volatile.**

3.3) (2 points) Identify the sequential logic circuit shown in the **Figure** below by labeling its inputs and outputs (where (1) and (2) are inputs and (3) and (4) are outputs). **SR-latch (1 = R, 2 = S, 3 = Q and 4 = $\overline{Q}$)**



3.4) (3 points) Identify the sequential logic circuit shown in the **Figure** below by labeling its inputs and outputs (where (1) and (2) are inputs and (3) and (4) are outputs). **D-latch (1 = D, 2 = CLK, 3 = Q and 4 = $\overline{Q}$)**



3.5) (3 points) Show how the designers eliminate the **forbidden state of the S-R latch** (show the <u>resulting latch with its circuit schematic and truth table</u>). **Find the JK-latch circuit and its truth table from the lecture slide**

3.6) (4 points) Identify the function of the logic circuit shown in the **Figure** below and briefly describe its function by creating its performance **truth table** (where *Clk* is the clock input and Q_0, Q_1, Q_2, and Q_3 are outputs). **Mod-14 counter (counts from 0000 (0) to 1101(13) and resets all the flip-flops)**